U. Roll No. | | | | | | | | | | | | | |

# Model Question Paper

**COURSE- B.Tech**

**DEPTT – Applied Sciences**

**YEAR: - 1st (Second SEM)**

**SUBJECT NAME: -Engineering Physics (BAS201)**

**TIME: 3.00 Hours** **TOTAL MARKS: 70**

**Note: Answer all questions. Draw neat sketch, wherever necessary. Suitably assume missing data, if any.**

## *Section A*

**Q1. Attempt all the parts.** **[2X7=14]**

| | | | |
|---|---|---|---|
| **a.** | What is Planck's law for black body radiation? | 2 | **[CO1], K1** |
| **b.** | What is skin depth? | 2 | **[CO2], K1** |
| **c.** | What is Rayleigh's criterion of resolution? | 2 | **[CO3], K2** |
| **d.** | Why Newton's rings are circular in nature? | 2 | **[CO3], K1** |
| **e.** | What is the acceptance angle and acceptance cone in an optical fiber? | 2 | **[CO4], K1** |
| **f.** | What is population inversion? | 2 | **[CO4], K1** |
| **g.** | What is the Meissner effect? | 2 | **[CO5], K1** |

## *Section B*

**Q2. Attempt any three of the following:** **[7X3=21]**

| | | | |
|---|---|---|---|
| **a.** | Discuss the Davison – Germer experiment in detail to demonstrate the wave nature of particle | 7 | **[CO1], K2** |
| **b.** | Write down the Maxwell equation in differential form for free space. Show that the velocity of plane electromagnetic waves in free space is given by $c = 1/\sqrt{\mu_0 \epsilon_0}$ $c = 1/\sqrt{\mu_0 \epsilon_0}$. | 7 | **[CO2], K3** |
| **c.** | Give the construction and theory of plane transmission grating. Explain and analyze the formation of spectra by it. | 7 | **[CO3], K2** |
| **d.** | What do you mean by the pulse broadening? Discuss dispersion in optical in detail. Give some applications of optical fiber | 7 | **[CO4], K2** |
| **e.** | Write some properties and applications of Nanomaterials. Describe Sol Gel method for synthesis of nano materials. | 7 | **[CO5], K2** |

## *Section C*

**Q3. Attempt any one part of the following:** **[7X1=7]**

| | | | |
|---|---|---|---|
| **a.** | Solve Schrodinger wave equation for a particle in one dimensional rigid box of side '*L*' and having potential energy (*V*) as follows:<br>$V(x) = \infty$ $V(x) = \infty$ for $x < 0 \; and \; x > L$ $x < 0 \; and \; x > L$<br>$V(x) = 0$ $V(x) = 0$ for $0 \le x \le L$ $0 \le x \le L$ | 7 | **[CO1], K3** |
| **b.** | A particle confined to move along x – axis has the wave function ψ = ax between x=0 & x=1.0 and ψ =0. Find the probability that a particle can be found between x= 0.35 to x= 0.45. | 7 | **[CO1], K3** |

**Q4. Attempt any one part of the following:** **[7X1=7]**

| | | | |
|---|---|---|---|
| **a.** | State and explain the poynting theorem for the flow of energy in electromagnetic waves. Also give the physical significance of the Poynting theorem. | 7 | **[CO2], K2** |

| | | | |
|---|---|---|---|
| **b.** | Assuming that all the energy from a 1000-watt lamp is reradiated uniformly; calculate the average value of intensities of electric and magnetic field of radiation at a distance of 2m from the lamp | 7 | **[CO2], K3** |

**Q5. Attempt any one part of the following:** **[7X1=7]**

| | | | |
|---|---|---|---|
| **a.** | What do you understand by resolving power of a grating? Derive the necessary expression for resolving power of grating. | 7 | **[CO3], K2** |
| **b.** | A parallel beam of light ($\lambda$ = *5890* $\mathring{A}$) is incident on a thin glass plate ($\mu$ = 1.5) such that the angle of refraction into the plate is $60^0$. Calculate the smallest thickness of the glass plate which will appear dark by reflection | 7 | **[CO3], K3** |

**Q6. Attempt any one part of the following:** **[7X1=7]**

| | | | |
|---|---|---|---|
| **a.** | Explain construction and working of He-Ne laser. Give some applications of lasers. | 7 | **[CO4], K2** |
| **b.** | What do you mean by attenuation in optical fiber? Discuss the important factors responsible for power loss in optical fiber.<br>A communication system uses 20 km fiber having a loss of 2.3 dB/km. Compute output power if input power is 400mW. | 7 | **[CO4], K3** |

**Q7. Attempt any one part of the following:** **[7X1=7]**

| | | | |
|---|---|---|---|
| **a.** | Write a short note on high temperature superconductors.<br>The superconducting transition temperature of lead is 7.26 K. The initial field at 0K is $64\times10^3$ A/m. Calculate the critical field at 5 K. | 7 | **[CO5], K3** |
| **b.** | Classify the superconductors? Write some properties and applications of superconductors. | 7 | **[CO5], K4** |